

# Quaternion-based Differentiator Adaptive Single-Gain Sliding Mode Controller

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**Abstract**—In this article, a differentiator-based controller for a fully actuated hexacopter unmanned aerial vehicle is presented, with a novel unit quaternion-based fixed-time differentiator for the attitude part, estimating the angular velocity, and angular position within a fixed time, despite the initial estimation error. A novel adaptive single-gain sliding mode controller is addressed to drive the linear and angular pose of the aircraft with practical finite-time convergence, robustness against external bounded perturbations under the Von Kármán Wind Model and uncertainties, chattering attenuation, and true assessment of the controller gains. Finally, simulations in the Simulink/Simscape Multibody demonstrate the feasibility of the proposed scheme.

**Index Terms**—Tilted-rotors hexacopter, adaptive sliding mode, perturbed environments, robust control.

## I. INTRODUCTION

The thesis research project is organized as follows: Section II presents the preliminaries, definitions, and lemmas, while Section III introduces the quaternion-based equations of motion and the Von Kármán Perturbation Model. The differentiator-based controller is given in Section IV while Section V unveils the results and their discussion. Finally, in Section VI the conclusions of the document are illustrated.

### A. Notation

$D_*\mathbf{a}$  represents element-wise operations, where  $D_*$  denotes a diagonal matrix of the vector  $*$ , and  $\mathbf{a}$  is another vector of the same size as  $*$ . Likewise, the functions absolute value, powers, sign, and sig functions are used as element-wise operations.

## II. PRELIMINARIES

This section presents the mathematical background for the quaternion-based controller and definitions and lemmas that support the Fixed-time Differentiator design.

### A. Quaternion algebra

Quaternions are considered hyper-complex numbers as a part of the group  $\mathbb{H}$ . For this paper we defined a quaternion as  $\mathbf{q} = q_0 + \vec{q} \in \mathbb{H}$ , consisting of 4 numbers,  $q_0$  often called as the scalar part  $q_0 \in \mathbb{R}$ , and the vector part  $\vec{q} = [q_1, q_2, q_3]^T \in \mathbb{R}^3$ .

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The product between two quaternions  $\mathbf{q}$  and  $\mathbf{p}$  using the  $\otimes$  symbol is defined as follows:

$$\mathbf{q} \otimes \mathbf{p} := \begin{bmatrix} q_0 & -q_1 & -q_2 & -q_3 \\ q_1 & q_0 & -q_3 & q_2 \\ q_2 & q_3 & q_0 & -q_1 \\ q_3 & -q_2 & q_1 & q_0 \end{bmatrix} \begin{pmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{pmatrix} \quad (1)$$

Then, the conjugate of a quaternion  $\mathbf{q}$  is defined as:

$$\mathbf{q}^* := q_0 - \vec{q} \quad (2)$$

The conjugate of a product between quaternions is

$$(\mathbf{q} \otimes \mathbf{p})^* = \mathbf{p}^* \otimes \mathbf{q}^* \quad (3)$$

In this paper, unit quaternions are used, one can calculate the unit quaternion, defined as the norm of  $\mathbf{q}$  as:

$$\|\mathbf{q}\|^2 := \mathbf{q} \otimes \mathbf{q}^* = q_0^2 + q_1^2 + q_2^2 + q_3^2 \quad (4)$$

One can calculate the axis-angle attitude representation of a given rotation from a quaternion  $\mathbf{q}$  with the logarithmic mapping.

$$\vec{\vartheta} = 2 \ln \mathbf{q} \quad (5)$$

$$\ln := \begin{cases} [0, 0, 0]^T & \text{if } \|\vec{q}\| = 0 \\ \frac{\vec{q}}{\|\vec{q}\|} \arccos q_0 & \text{if } \|\vec{q}\| \neq 0 \end{cases} \quad (6)$$

where  $\vec{\vartheta} \in \mathbb{R}^3$  is the axis-angle attitude representation.

### B. Definitions and lemmas

This section presents the definitions and lemmas that support the design of the differentiator design. Consider the nonlinear system.

$$\dot{x}(t) = f(x, t, u), x_0 = x(t_0) \quad (7)$$

where  $x$  is the state,  $x_0$  is the initial condition of  $x$ ,  $f(x, t, u)$  is a nonlinear function and  $u$  is the control input.

**Definition 1** [7] The origin is globally finite-time stable if it is globally asymptotically stable and any solution achieves the equilibrium at some finite time given by a settling time function  $T_m(x_0)$  i.e.  $x(x_0, t) = 0 \forall t \geq t_0 + T_m(x_0)$ .

**Definition 2** [8] The solutions are said to be practical finite-time stable if for all  $x(t_0) = x_0$  there is a sufficiently small  $k > 0$  that satisfies  $|x(t)| < k \forall t \geq t_0 + T_m$  with  $T_m(k, x_0) < \infty$ .

**Definition 3.** [5] System is uniformly convergent if all the trajectories  $x(t)$  are bounded by  $|x(t)| \leq m \forall x_0 \in \mathbb{R}^n$  and  $\forall t \geq T_k$  with  $m \in \mathbb{R}^+$  small enough, and  $T_k > 0$ .

**Definition 4** [7] [5] System (1) is fixed-time stable in the origin  $x = 0$  if

- It is globally finite-time stable and uniformly convergent
- There is a settling time  $T_m(x_0)$  upper bounded by an explicitly indicated constant  $T_k > 0$  such that  $T_m(x_0) \leq T_k \implies x(t) = 0 \forall t \geq t_0 + T_k$  for any initial condition  $x_0$ .

**LEMMA 1.** [5] If a set of real-valued continuous functions  $V_1$  and  $V_2$ , with  $V_1$  positive definite, are homogeneous with respect to dilation  $\omega$  of degree  $\iota_1 > 0$  and  $\iota_2 > 0$ , the inequality

$$\left[ \min_{\{z: V_q(z)=1\}} V_2(z) \right] [V_1(x)^{\frac{\iota_2}{\iota_1}}] \leq V_2(x) \leq \left[ \max_{\{z: V_q(z)=1\}} V_2(z) \right] [V_1(x)^{\frac{\iota_2}{\iota_1}}] \quad (8)$$

is true  $\forall x \in \mathbb{R}^n$ .

**LEMMA 2.** Consider system (1) is also in the form of a homogeneous continuous function vector field  $v_f$  of degree  $q$  concerning weights  $p$ . The origin of  $v_f$  is finite-time stable if and only if the origin is asymptotically stable and  $q < 0$ . Then, there is a continuous positive definite and homogeneous function  $\mathcal{V}(x) : \mathbb{R}^n \rightarrow \mathbb{R}$  of degree  $k$  with  $k > \max\{-q, 0\}$  such that its derivative:

$$\dot{\mathcal{V}}(x) \leq -\varepsilon \mathcal{V}f(x)^{\frac{k+q}{k}} \quad (9)$$

is continuous negative definite for any  $\varepsilon > 0$  and homogeneous of degree  $k + q > 0$ .

**LEMMA 3.** [8] Consider system (1) and assume the existence of a continuous function  $V(x)$ , scalars  $\mathcal{F} > 0, 0 < \varrho < 1$ , and  $0 < \kappa < 1$  such that

$$\dot{V}(x) \leq \mathcal{F}V(x)^\varrho + \kappa \quad (10)$$

Then, system (1) is practical finite-time stable and its trajectories are bounded in finite time as

$$\lim_{\vartheta \rightarrow \vartheta_0} x \in \left( V(x)^\varrho \leq \frac{\kappa}{(1-\vartheta)\mathcal{F}} \right) \quad (11)$$

with  $0 < \vartheta < 1$  and  $0 < \vartheta_0 < 1$ . Such trajectories are achieved at a time bounded by:

$$T_m(x_0) \leq \frac{V(x_0)^{1-\varrho}}{\mathcal{F}\vartheta_0(1-\varrho)} \quad (12)$$

**LEMMA 4.** [5] Consider a continuous vector field  $v_g$ . Then, system (1) is said to be uniformly convergent for initial condition  $x_0$  if it is globally asymptotically stable at the origin and  $v_g$  is homogeneous of degree  $v > 0$ .

**LEMMA 5.** [7] Consider a set  $E$  and a continuous radially unbounded function  $\mathcal{V} : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{0\}$  that fulfills: -  $\mathcal{V}(x) = 0 \implies x \in E$ . - Any solution of (1) satisfies

$$\dot{\mathcal{V}} \leq -(\gamma_1 V(x)^{c_1} + \gamma_2 V(x)^{c_2})^{c_3} \quad (13)$$

with  $\gamma_1, \gamma_2, c_1, c_2, c_3 > 0$ ,  $c_1 c_3 < 1$ , and  $c_2 c_3 > 1$ . Then, set  $E \in \mathbb{R}^n$  is globally fixed-time attractive for and the settling function  $T_m(x_0)$  is bounded by:

$$T_m(x_0) \leq \frac{1}{\gamma_1^{c_3}(1-c_1 c_3)} + \frac{1}{\gamma_2^{c_3}(c_2 c_3 - 1)} \quad (14)$$

for all  $x_0 \in \mathbb{R}^n$

### III. HEXA-ROTOR DYNAMICS

#### A. FAH Dynamics

This section addresses the quaternion-based equations of motion [1] that govern the rotorcraft dynamics and the turbulent wind modeling.

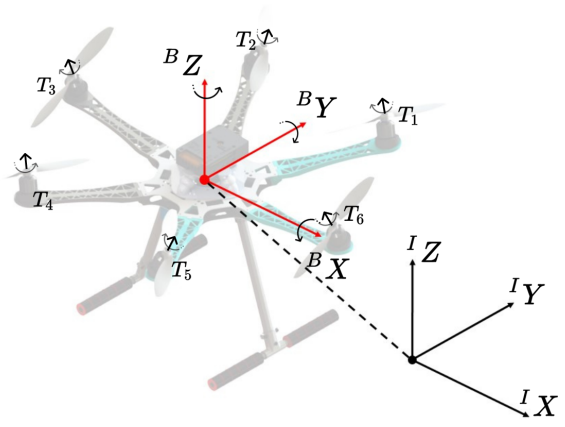


Fig. 1. Reference frames for the FAH.

Then, applying Newton's equations of motion for a rigid body with a unit quaternion we get:

$$\begin{bmatrix} \ddot{\gamma} \\ \dot{q} \\ \dot{\omega} \end{bmatrix} = \begin{bmatrix} m^{-1} \mathbf{F}^I + \mathbf{g} + \boldsymbol{\delta} \\ \frac{1}{2} \mathbf{q} \otimes \boldsymbol{\omega} \\ J^{-1}(\boldsymbol{\tau} - \boldsymbol{\omega} \times J\boldsymbol{\omega}) \end{bmatrix} \quad (15)$$

Where  $\gamma, \dot{\gamma} \in \mathbb{R}^3$ , and  $\ddot{\gamma}$  denotes the inertial position, velocity, and acceleration, respectively. The FAH attitude is represented as a unit quaternion  $\mathbf{q}$  and the angular velocities  $\boldsymbol{\omega} \in \mathbb{R}^3$ . The thrust force is defined as  $\mathbf{F}^I = [F_x, F_y, F_z]^T$ , where lateral forces are applied due to the tilted rotors, and the gravity vector  $\mathbf{g} = [0, 0, g]^T$ .  $\mathbf{F}^I$  is calculated through:

$$\mathbf{F}^I = \mathbf{q} \otimes \mathbf{F}^B \otimes \mathbf{q}^* \quad (16)$$

The moments of inertia are condensed in matrix  $\mathbf{J} = \text{diag}[J_{xx}, J_{yy}, J_{zz}]$ , and the vector  $\boldsymbol{\delta}$  stands for the wind drag forces, expressed as:

$$\boldsymbol{\delta} = -\frac{1}{2} \rho D_{\boldsymbol{\eta}} D_{\mathbf{A}} (\dot{\gamma} - \mathbf{v}_w)^2 \text{sign}(\dot{\gamma} - \mathbf{v}_w) \quad (17)$$

where  $\rho$  is the air density,  $\boldsymbol{\eta} \in \mathbb{R}^3$  are the drag coefficients of the FAH,  $\mathbf{A} \in \mathbb{R}^3$  denotes the transversal area per axis. Then,  $\mathbf{v}_w \in \mathbb{R}^3$  is calculated by:

$$\mathbf{v}_w = \bar{\mathbf{v}}_w + \Delta \quad (18)$$

Where  $\bar{\mathbf{v}}_w$  is the wind mean velocity and  $\Delta$  are the winds generated under the Von Kármán wind model.

### IV. CONTROLLER DESIGN

This section presents the proposed differentiator-based controller composed of the Fixed-Time Quaternion-based Differentiator(FxTQDiff), and an Adaptive Single-Gain Sliding Mode Controller (ASGSM). The FxTQDiff provides full

state estimation within a fixed time no matter the initial conditions of the differentiator. Consider that there are two differentiator designs, one for the positions and attitude subsystem. The differentiator-based controller scheme is visualized in Fig. 2.

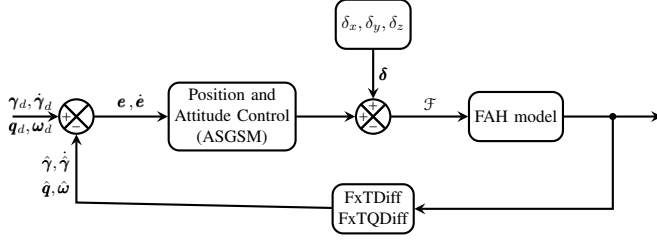


Fig. 2. FxTDiff-ASGSMC scheme.

*Assumption 4.1:* Only linear and angular position vectors are available for design from the FAH as  $\gamma$ , and  $q$ , respectively. Additionally, linear and angular velocities of the rotorcraft are upper-bounded by  $\|\dot{\gamma}\| \leq \Gamma$ , and  $\|\dot{q}\| \leq \Omega$ , respectively with  $\Gamma, \Omega \in \mathbb{R}^+$ .

*Assumption 4.2:* External disturbances given by  $\delta$  are continuously differentiable, and upper-bounded by  $\|\delta\| \leq \delta_{\max}$ , with  $\delta_{\max} > 0$ .

#### A. Fixed-Time Differentiator

Then, consider the following single-input single-output second-order nonlinear system:

$$\begin{aligned} \dot{\zeta}_1 &= \zeta_2 \\ \dot{\zeta}_2 &= f(\zeta_1, \zeta_2) + g(\zeta_1, \zeta_2)u \\ \gamma_1 &= \zeta_1 \end{aligned} \quad (19)$$

where  $\zeta_1$ , and  $\zeta_2$  denotes the system states,  $f(\zeta_1, \zeta_2)$  a continuous differentiable function,  $|g(\zeta_1, \zeta_2)| > 0$ , and  $u$  is the control input. Term  $\gamma$  and its first derivative can be obtained with the second-order fixed-time differentiator [3].

$$\dot{\hat{\gamma}}_1 = \hat{\gamma}_2 + G_1 \lceil \tilde{\gamma} \rceil^{\frac{\alpha_l+1}{2}} + Q_1 \lceil \tilde{\gamma} \rceil^{\frac{\beta_l+1}{2}} \quad (20)$$

$$\dot{\hat{\gamma}}_2 = G_2 \lceil \tilde{\gamma} \rceil^{\alpha_l} + Q_2 \lceil \tilde{\gamma} \rceil^{\beta_l} \quad (21)$$

where  $\lceil \tilde{\gamma} \rceil^a = |\tilde{\gamma}|^a \text{sign}(\tilde{\gamma})$  and  $\tilde{\gamma} = \gamma - \hat{\gamma}$  denotes the estimation error. Gains  $G_1, Q_1, G_2, Q_2 \in \mathbb{R}^+$ ,  $0 < \alpha_l < 1$ ,  $\beta_l > 1$ .

#### B. Position Controller

Defining the system errors as:

$$\begin{aligned} e &= \gamma_d - \hat{\gamma} \\ \dot{e} &= \dot{\gamma}_d - \dot{\hat{\gamma}} \end{aligned} \quad (22)$$

Where the desired positions, velocities, and accelerations are given by  $\gamma_d, \dot{\gamma}_d, \ddot{\gamma}_d$ , respectively. The Non-Singular Fast Terminal Sliding Surface [6] is defined as follows:

$$\sigma = e + D_{\xi_1} D_{\text{sign}(e)} |e|^{\lambda_1} + D_{\xi_2} D_{\text{sign}(\dot{e})} |\dot{e}|^{\lambda_2} \quad (23)$$

where the sliding surface design parameters  $\xi_1, \xi_2, \lambda_1, \lambda_2 \in \mathbb{R}^+$ , satisfying  $1 < \lambda_{2_i} < 2$  and  $\lambda_{1_i} > \lambda_2$ . This guarantees

finite time convergence and provides non-singularity. Then the derivative of the sliding surface results in:

$$\dot{\sigma} = \dot{e} + D_{\lambda_1} D_{\xi_1} D_{|e|^{\lambda_1-1}} \dot{e} + D_{\lambda_2} D_{\xi_2} D_{|\dot{e}|^{\lambda_2-1}} \ddot{e} \quad (24)$$

$$\begin{aligned} \dot{\sigma} &= \dot{e} + D_{\lambda_1} D_{\xi_1} D_{|e|^{\lambda_1-1}} \dot{e} \\ &\quad + D_{\lambda_2} D_{\xi_2} D_{|\dot{e}|^{\lambda_2-1}} (\ddot{\gamma}_d - \ddot{\gamma}) \end{aligned} \quad (25)$$

Then, from sliding dynamics (25), one gets

$$\ddot{\gamma} = m^{-1} F^T + g \quad (26)$$

the following feedback controller is proposed:

$$u = m(-g + \varphi + \ddot{\gamma}_d - u_{\text{aux}}) \quad (27)$$

where,

$$\varphi = D_{\lambda_2}^{-1} D_{\xi_2}^{-1} D_{\text{sign}(\dot{e})} D_{|\dot{e}|^{2-\lambda_2}} (1 + D_{\lambda_1} D_{\xi_1} |e|^{\lambda_1-1}) \quad (28)$$

Finally, the  $u_{\text{aux}}$  incorporates the single gain method (AS-GSM), defined as:

$$u_{\text{aux}} = -2D_K D_{\text{sign}(\sigma)} |\sigma|^{1/2} - 2D_\epsilon D_K \sigma \quad (29)$$

Where  $0 < \epsilon_i < 1$  is a tuning parameter and  $K$  is the adaptive gain, obtained by integrating the following equation:

$$\dot{K} = D_\alpha^{1/2} |\sigma|^{1/2} - D_\beta^{1/2} K^2 \quad (30)$$

Where  $\alpha_i > 0$  and  $\beta_i > 0$  are the design parameters that define the sliding surface's precision and the system's control effort, respectively.

#### C. Fixed-Time Quaternion-based Differentiator

Analogous to the position differentiator-based controller, consider the single-input single-output second-order nonlinear system.

$$\begin{aligned} \dot{\chi}_1 &= \chi_2 \\ \dot{\chi}_2 &= f(\chi_1, \chi_2) + g(\chi_1, \chi_2)u \end{aligned} \quad (31)$$

Then, the following Fixed-time Quaternion-based Differentiator (FxTQD) is presented as:

$$\begin{aligned} \dot{\hat{q}} &= \frac{1}{2} \hat{q} \otimes \tilde{q} \otimes \left[ \hat{\omega} + \epsilon_1 \Gamma_q^{-1} \lceil \tilde{q}_e \rceil^{\frac{\alpha_q+1}{2}} \right. \\ &\quad \left. + \mu_1 \Gamma_q^{-1} \lceil \tilde{q}_e \rceil^{\frac{\beta_q+1}{2}} \right] \otimes \tilde{q}^* \end{aligned} \quad (32)$$

$$\dot{\hat{\omega}} = \epsilon_2 \Gamma_q^{-1} \lceil \tilde{q}_e \rceil^{\alpha_q} + \mu_2 \Gamma_q^{-1} \lceil \tilde{q}_e \rceil^{\beta_q} \quad (33)$$

where the unit quaternion estimation error is defined as  $\tilde{q} := \hat{q}^* \otimes q$ , and  $\tilde{q}_e$  is the vector part of the error estimation quaternion. Gains  $\epsilon_1, \epsilon_2, \mu_1, \mu_2 \in \mathbb{R}^+$ ,  $0 < \alpha_q < 1$ ,  $\beta_q > 1$ .  $\Gamma_q^{-1}$  is denoted as

$$\Gamma_q^{-1} = \left( \frac{1}{2} \begin{bmatrix} \tilde{q}_0 & -\tilde{q}_3 & \tilde{q}_2 \\ \tilde{q}_3 & \tilde{q}_0 & -\tilde{q}_1 \\ -\tilde{q}_2 & \tilde{q}_1 & \tilde{q}_0 \end{bmatrix} \right)^{-1} \quad (34)$$

To demonstrate the stability of the proposed FxTQD define the estimations errors as:

$$\tilde{q} := \hat{q}^* \otimes q \quad (35)$$

$$\tilde{\omega} = \omega - \hat{\omega} \quad (36)$$

One can define the estimation error system based on equations (32), (33), (35), and (36), as well as the presented quaternion algebra.

$$\dot{\tilde{q}} = \frac{1}{2} \tilde{q} \otimes \left[ \tilde{\omega} - \epsilon_1 \Gamma_q^{-1} \text{sig}^{\frac{\alpha+1}{2}}[e_q] - \mu_1 \Gamma_q^{-1} \text{sig}^{\frac{\beta+1}{2}}[e_q] \right] \quad (37)$$

$$\dot{\tilde{\omega}} = -\epsilon_2 \Gamma_q^{-1} \text{sig}^\alpha[e_q] - \mu_2 \Gamma_q^{-1} \text{sig}^\beta[e_q] \quad (38)$$

From (37) take the vector part to further analysis as  $\tilde{q}_v$ , then

$$\dot{\tilde{q}}_v = \Gamma_q \left[ \tilde{\omega} - \epsilon_1 \Gamma_q^{-1} \text{sig}^{\frac{\alpha+1}{2}}[e_q] - \mu_1 \Gamma_q^{-1} \text{sig}^{\frac{\beta+1}{2}}[e_q] \right] \quad (39)$$

Split (39) and (38) into subsystems

$$Z_\alpha = \begin{cases} \dot{\tilde{q}}_v = \Gamma_q^{-1} \tilde{\omega} - \epsilon_1 \text{sig}^{\frac{\alpha+1}{2}}[e_q] \\ \dot{\tilde{\omega}} = -\epsilon_2 \Gamma_q^{-1} \text{sig}^\alpha[e_q] \end{cases} \quad (40)$$

$$Z_\beta = \begin{cases} \dot{\tilde{q}}_v = \Gamma_q^{-1} \tilde{\omega} - \mu_1 \text{sig}^{\frac{\beta+1}{2}}[e_q] \\ \dot{\tilde{\omega}} = -\mu_2 \Gamma_q^{-1} \text{sig}^\beta[e_q] \end{cases} \quad (41)$$

For the first subsystem  $Z_\alpha$ , based on the change of coordinates proposed in [4]

$$\xi = [\xi_1, \xi_2] \quad (42)$$

where  $\xi_1 = |\tilde{q}_v|^{(\alpha+1)/2} \text{sign}(\tilde{q}_v)$ ,  $\xi_2 = \tilde{\omega}$ . Then, deriving (42) one gets

$$\dot{\xi} = \begin{bmatrix} \dot{\xi}_1 \\ \dot{\xi}_2 \end{bmatrix} = \Xi \begin{bmatrix} -\frac{\alpha+1}{2} \mu_1 I_{3 \times 3} & \frac{\alpha+1}{2} \Gamma_q \\ -\Gamma_q^{-1} \mu_2 & 0_{3 \times 3} \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix} \quad (43)$$

where  $\Xi = \text{diag}(|\tilde{q}_v|^{(\alpha-1)/2}, |\tilde{q}_v|^{(\alpha-1)/2}) \in \mathbb{R}^{6 \times 6}$ . Then, now based on the coordinate transformation of [2]  $T$  defined as

$$T = \begin{bmatrix} \Gamma_q^{-1} & 0_{3 \times 3} \\ 0_{3 \times 3} & I_{3 \times 3} \end{bmatrix} \quad (44)$$

then, define  $\bar{\xi} = T\xi$ , one obtains

$$\begin{aligned} \dot{\bar{\xi}} &= \dot{T}\xi + T\dot{\xi} = \dot{T}T^{-1}\bar{\xi} + \Xi T A \xi \\ &= \dot{T}T^{-1}\bar{\xi} + \Xi T A T^{-1}\bar{\xi} \end{aligned} \quad (45)$$

where

$$\dot{T}T^{-1} = \begin{bmatrix} \dot{\Gamma}_q^{-1} \Gamma_q & 0_{3 \times 3} \\ 0_{3 \times 3} & 0_{3 \times 3} \end{bmatrix} := \bar{T} \quad (46)$$

$$T A T^{-1} = \begin{bmatrix} -\frac{\alpha+1}{2} \mu_1 I_{3 \times 3} & \frac{\alpha+1}{2} I_{3 \times 3} \\ -\mu_2 I_{3 \times 3} & 0_{3 \times 3} \end{bmatrix} := \bar{A} \quad (47)$$

then,

$$\dot{\bar{\xi}} = \bar{T}\bar{\xi} + \Xi \bar{A}\bar{\xi} \quad (48)$$

Therefore, the following Lyapunov Candidate function is proposed

$$V_\alpha = \bar{\xi}^T R \bar{\xi} \quad (49)$$

Where  $R$  is a positive definite symmetric matrix satisfying the Lyapunov equation

$$R\bar{A} + \bar{A}^T R = -\varepsilon I_{6 \times 6} \quad (50)$$

where  $\varepsilon > 0$ , and  $\bar{A}$  is Hurwitz. Then, deriving the Candidate Lyapunov function one gets

$$\dot{V}_\alpha = \dot{\bar{\xi}}^T R \bar{\xi} + \bar{\xi}^T R \dot{\bar{\xi}} \quad (51)$$

$$= \bar{\xi}^T [\bar{T}^T + \bar{A}^T \Xi^T] R \bar{\xi} + \bar{\xi}^T R [\bar{T} + \Xi \bar{A}] \bar{\xi} \quad (52)$$

$$= \bar{\xi}^T \bar{T}^T R \bar{\xi} + \bar{\xi}^T \bar{A}^T \Xi^T R \bar{\xi} + \bar{\xi}^T R \bar{T} \bar{\xi} + \bar{\xi}^T R \Xi \bar{A} \bar{\xi} \quad (53)$$

#### D. Attitude Controller

To generate the FxTQD-based controller for the attitude, consider the quaternion error as:

$$q_e = \hat{q}^* \otimes q_d \quad (54)$$

Where  $q_d$  is the desired attitude position in terms of a unit quaternion. Then, with the logarithm mapping, the quaternion error is transformed into the axis-angle error:

$$e = 2\ln|q_e| \quad (55)$$

$$\dot{e} = \omega_d - \hat{\omega} \quad (56)$$

$$\ddot{e} = \dot{\omega}_d - \dot{\hat{\omega}} \quad (57)$$

Analogous to the positioning subsystem with the NSFTS one gets

$$\sigma_q = e + D_{\rho_1} \text{sig}^{\varpi_1}[e] + D_{\rho_2} \text{sig}^{\varpi_2}[\dot{e}] \quad (58)$$

where  $\rho_1, \rho_2 \in \mathbb{R}^+$ ,  $\varpi_1 > \rho_2$ , and  $1 < \varpi_2 < 2$ . Therefore, one can get the sliding dynamics as:

$$\dot{\sigma}_q = \dot{e} + D_{\rho_1} D_{\varpi_1} D_{|e|^{\varpi_1-1}} \dot{e} + D_{\rho_2} D_{\varpi_2} D_{|\dot{e}|^{\varpi_2-1}} \ddot{e} \quad (59)$$

second-order derivatives are:

$$\ddot{e} = \dot{\omega}_d - J^{-1}(\tau - \omega \times \omega J) \quad (60)$$

Finally, selecting the control input  $\tau$  with feedback control substituting 60 in 59 as:

$$\tau = J \left[ \dot{\omega}_d - J^{-1} \omega \times J \omega + \kappa - u_{\text{aux}} \right] \quad (61)$$

where,

$$\kappa = D_{\varpi_2}^{-1} D_{\rho_2}^{-1} D_{\text{sign}(\dot{e})} D_{|\dot{e}|^{2-\varpi_2}} \left( 1 + D_{\varpi_1} D_{\varpi_1} |e|^{\varpi_1-1} \right) \quad (62)$$

Analogous to the positioning subsystem, the  $u_{\text{aux}}$  is described as in 29.

## V. RESULTS

This section presents numerical simulation utilizing the Multibody simulator from Matlab, Simulink, and Sismcape, representing a closer dynamics environment. The performance of the FxTQDiff-ASGSM controller approach in perturbed environments is showcased, while the differentiator's initial estimations are left at 0. The rotorcraft weight is set as 1.7 kg, while the inertias are  $I_x = I_y = 0.03$ , and  $I_z = 0.05$ .

The FAH flights consist of a 60 s circular trajectory of 4 [m], and a constant value of 2 [m] function for the z-axis, in addition, a continuous rotation in the yaw angle is

TABLE I  
ASGSM GAINS

| Gain        | $x$  | $y$  | $z$   | $\phi$ | $\theta$ | $\psi$ |
|-------------|------|------|-------|--------|----------|--------|
| $\xi_1$     | 2    | 2    | 1.5   | 1      | 1        | 1      |
| $\xi_2$     | 1    | 1    | 2     | 0.5    | 0.5      | 0.5    |
| $\lambda_1$ | 1.5  | 1.5  | 1.75  | 2      | 2        | 1.75   |
| $\lambda_2$ | 1.2  | 1.2  | 1.25  | 1.2    | 1.2      | 1.2    |
| $\alpha$    | 1    | 1    | 0.625 | 0.1    | 0.1      | 0.1    |
| $\beta$     | 1.25 | 1.25 | 0.1   | 0.05   | 0.05     | 0.25   |
| $\epsilon$  | 0.4  | 0.4  | 0.4   | 0.4    | 0.4      | 0.4    |

indicated. The equation that governs said desired trajectories is as follows:

$$\begin{aligned}
 x_d &= \begin{cases} 0, & t < 5(s) \\ -2 + 2 \cos(0.15(t-5)), & t \geq 5(s) \end{cases} \\
 y_d &= \begin{cases} 0, & t < 5(s) \\ 2 \sin(0.15(t-5)), & t \geq 5(s) \end{cases} \\
 \psi_d &= \begin{cases} 0, & t < 0(s) \\ 2\pi(t-5)/(p-5), & t \geq 5(s) \end{cases}
 \end{aligned} \quad (63)$$

where  $p = 40\frac{\pi}{3} + 5$ . Initial conditions of the system are  $\gamma = [0, 0, 0]^T$ , and  $\mathbf{q} = [0.99950.02520.0146 - 0.0144]^T$ . The controller gains are presented in table I. The FxtDiff was set with gains  $G_1 = Q_1 = 5$ ,  $G_2 = Q_2 = 6$ ,  $\alpha_l = 0.6$ , and  $\beta_l = 1.2$ . While for the FxTQDiff as  $\epsilon_1 = \mu_1 = 5$ ,  $\epsilon_2 = \mu_2 = 6$ ,  $\alpha_q = 0.6$ ,  $\beta_q = 1.2$ . External perturbations under the Von Kármán scheme start at  $t = 5s$ .

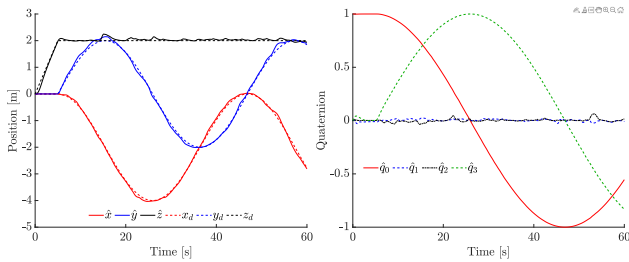


Fig. 3. Estimated states vs desired.

In (3) the desired vs estimated states plots are visualized, and one can infer that the systems track the desired positions accordingly, even with external perturbations as drag forces. The quaternion represents the estimate acquired from the proposed algorithm, and one can see how the real  $q_0$  and  $q_3$  changes in continuous functions, even though the drone made a complete turn in the yaw angle, this enforces the advantage of continuous angle rotations as well as no gimbal lock, opposed to traditional Euler angles.

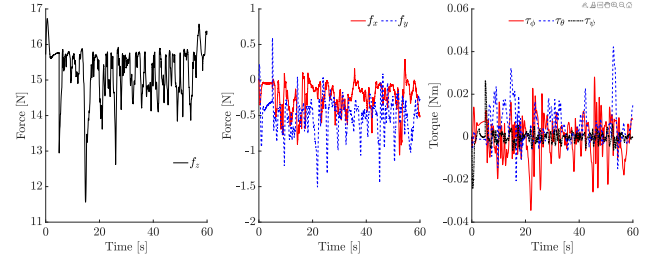


Fig. 4. Control inputs.

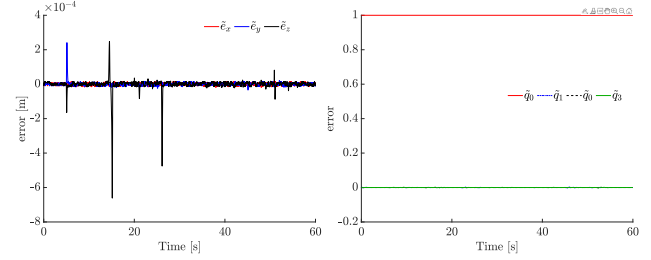


Fig. 5. Estimation errors.

In Fig. (4) the control inputs are showcased, the thrust denoted as  $f_z$  changes to take action against external perturbations, as well as the horizontal forces, the thrust stays in between 17 and 12 [N], which is a reasonable control force for a rotorcraft of 1.7 kg. The torques stay in between  $\pm 0.04$ , which against, are reasonable for the rotorcraft size and weight.

In addition, the estimation error of both of the differentiators is visualized in Fig. (5), for the linear positions with minimal peaks considering the dimension of the y-axis, while for the attitude estimation error, the aforementioned unit quaternion estimation error stays close to  $[1, 0, 0, 0]^T$ , which denotes that there is a rotation approximately of  $[0, 0, 0]^T$  between the actual and estimated quaternion.

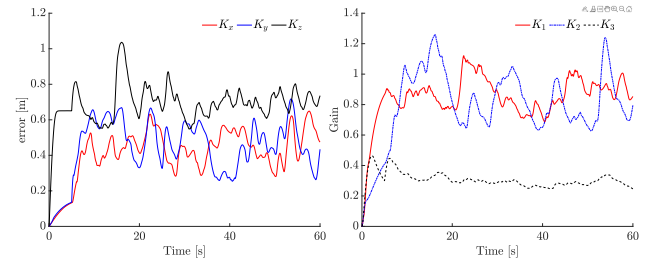


Fig. 6. Adaptive gains.

Finally, the adaptive gains of the proposed single gain controller are observed in Fig. (6), where smooth changes are presented. At the start of the simulation where there are no perturbations, minimal changes are perceived, but once the disturbances are added till the end of the simulation, gain changes according to the magnitude of said perturbations, said estimations were obtained in a fixed time no matter the initial conditions.

## VI. CONCLUSIONS

In this work, an adaptive nonsingular fast terminal sliding mode controller was designed to control the position and attitude of a Fully-Actuated Hexarotor. A Fixed-time Differentiator was used to calculate the first derivative, as well as a novel quaternion-based differentiator encapsulating quaternion math was employed to obtain estimated angular velocities. Lyapunov Analysis is being constructed, but a general idea of the definitions, lemmas, and coordinated transformation has been defined. Finally, simulations in the Matlab/Simulink/Simscape environment indicated the capabilities of the rotorcraft to follow desired trajectories with the existence of external perturbations and demonstrated the robustness and effectiveness of the proposal.

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